$x = a\cos\theta, y = a\sin\theta,\ 0 \le \theta \le 2\pi$ are the parametric equations of the circle $x^2 + y^2 = a^2$.

Conversely, if $x = a\cos\theta,\ y = a\sin\theta,\ 0 \le \theta \le 2\pi,$

then, $\frac{x}{a} = \cos\theta, \frac{y}{a} = \sin\theta$.

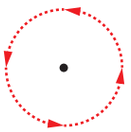
Squaring and adding, we get,

$$\frac{x^2}{a^2} + \frac{y^2}{a^2} = \cos^2\theta + \sin^2\theta = 1.$$

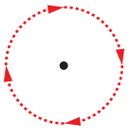
Thus $x^2 + y^2 = a^2$ yields the equation to circle with centre $(0,0)$ and radius $a$ units.

**Note**

(1) $x = a\cos t, y = a\sin t,\ 0 \le t \le 2\pi$ also represents the same parametric equations of circle $x^2 + y^2 = a^2$,
$t$ increasing in anticlockwise direction.


Fig. 5.45

(2) $x = a\sin t, y = a\cos t, 0 \le t \le 2\pi$ also represents the same parametric equations of circle $x^2 + y^2 = a^2$,
$t$ increasing in clockwise direction.


Fig. 5.46

### (ii) Parametric form of the parabola $y^2 = 4ax$

Let $P(x_1, y_1)$ be a point on the parabola

$$y_1^2 = 4ax_1$$

$$(y_1)(y_1) = (2a)(2x_1)$$

$$\frac{y_1}{2a} = \frac{2x_1}{y_1} = t \quad (-\infty < t < \infty) \text{ say}$$

$$y_1 = 2at,\ 2x_1 = y_1 t$$

$$2x_1 = 2at(t)$$

$$x_1 = at^2$$

Parametric form of $y^2 = 4ax$ is $x = at^2, y = 2at, -\infty < t < \infty$.

Conversely if $x = at^2$ and $y = 2at, -\infty < t < \infty$, then eliminating '$t$' between these equations we get $y^2 = 4ax$.

### (iii) Parametric form of the Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Let $P$ be any point on the ellipse. Let the ordinate $MP$ meet the auxiliary circle at $Q$.

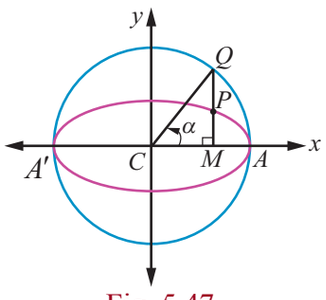


Fig. 5.47

Let $\angle ACQ = \alpha$

$\therefore\ CM = a\cos\alpha,\ MQ = a\sin\alpha$

and $Q(a\cos\alpha, a\sin\alpha)$

Now $x$-coordinate of $P$ is $a\cos\alpha$. If its $y$-coordinate is $y'$, then $P(a\cos\alpha, y')$ lies on

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$



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